

Luxury Cosmetics Pop-up Events Performance Analysis (2024–2025)

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INTRODUCTION

Luxury cosmetic brands increasingly rely on pop-up retail events to engage customers, test markets, and generate short-term sales across premium locations such as airport duty-free zones, luxury malls, high streets, art and design districts, and department store atriums. This project analyzes pop-up event performance between February 2024 and August 2025 across 24 luxury brands, 26 cities, and 5 global regions.

BUSINESS PROBLEM & OBJECTIVE

Pop-up events are expensive to run, particularly in the luxury segment. Brands often assume that airport duty-free locations deliver better conversions due to high footfall. This analysis evaluates whether airport pop-ups truly outperform non-airport locations in conversion efficiency and revenue generation.

HYPOTHESIS & KPI

Null Hypothesis: No performance difference exists between airport and non-airport pop-ups.

Alternative Hypothesis: Non-airport pop-ups have higher conversion efficiency.

Key KPI: $\text{Conversion Rate} = \text{Units Sold} \div \text{Average Daily Footfall}$.

DATA PREPARATION & METHODOLOGY

The dataset was sourced from Kaggle and cleaned using Excel, PostgreSQL, Tableau Public, and DBeaver. Data refinement included handling missing values, standardizing location types, and creating new metrics such as total revenue, weighted conversion rate, and location grouping.

STATISTICAL ANALYSIS & OUTLIERS

Key metrics such as total revenue, average revenue, and conversion rates were calculated. Outliers were identified but retained to preserve real-world luxury market behavior.

AIRPORT VS NON-AIRPORT ANALYSIS

Non-airport locations outperform airport duty-free locations in weighted conversion rates. Although airports contribute around 20% of revenue, nearly 80% comes from non-airport pop-ups. Art districts, luxury malls, and department store atriums show superior performance.

PRODUCT & CITY INSIGHTS

Fragrances generate nearly one-third of total revenue. Airport success is city-specific, with locations such as Paris and Lima performing better than others.

CONCLUSION

The hypothesis is accepted. Non-airport luxury pop-ups convert footfall into sales more effectively than airport duty-free events.

RECOMMENDATIONS

Prioritize luxury malls and art districts.
Use airports mainly for brand visibility.
Focus airport product mix on fragrances.
Optimize event duration based on performance.

Analysis done Using following queries-

Tools - Spreadsheets, PostgreSQL, Dbeaver, Tableau, Chatgpt, Canva , Google slides

QUERIES FOR AIRPORT VS NON-AIRPORT LOCATION TYPE ANALYSIS

1) How many events are Airport vs Non-Airport?

```
SELECT CASE WHEN
```

```
location_type = 'Airport Duty-Free' THEN 'Airport' ELSE 'Non-Airport' END AS
```

```
location_group, COUNT(*) AS total_events
```

```
FROM challan_test1
```

```
GROUP BY 1;
```

Answer: Airport = **434**,

Non-Airport = **1,699**

2) What % of all events are Airport Duty-Free?

```

SELECT
ROUND(100.0 * SUM(CASE WHEN
location_type='Airport Duty-Free' THEN 1 ELSE 0 END) / COUNT(*),
2) AS airport_event_pct

FROM challan_test1;

```

Answer: 20.35%

3) What is the weighted conversion rate (total units / total footfall) for Airport vs Non-Airport?

```

SELECT
CASE WHEN location_type = 'Airport Duty-Free' THEN 'Airport' ELSE 'Non-Airport' END
AS location_group,
ROUND(SUM(units_sold) * 1.0 / NULLIF(SUM(avg_daily_footfall), 0), 4) AS
conversion_rate
FROM challan_test1
GROUP BY 1;

```

Answer: Airport = 1.2391, Non-Airport = 1.4181 correct

4) What is the average conversion rate (avg of event-level conversion) for Airport vs Non-Airport?

```

SELECT
CASE WHEN location_type = 'Airport Duty-Free' THEN 'Airport'
ELSE 'Non-Airport' END AS location_group,
ROUND(AVG(units_sold * 1.0 / NULLIF(avg_daily_footfall, 0)), 4)
AS avg_conversion_rate
FROM challan_test1
GROUP BY 1;

```

Answer: Airport = 1.3625, Non-Airport = 1.6745 correct

5) What is the median conversion rate for Airport vs Non-Airport? (PostgreSQL)

```

SELECT
    CASE WHEN location_type = 'Airport Duty-
Free' THEN 'Airport' ELSE 'Non-Airport' END AS location_group,
ROUND(
    PERCENTILE_CONT(0.5)
    WITHIN GROUP (
        ORDER BY (units_sold * 1.0 / NULLIF(avg_daily_footfall,0))
    )::numeric,
    4
) AS median_conversion_rate
FROM challan_test1
GROUP BY 1;

```

Answer: Airport = **1.1896**, Non-Airport = **1.3804** correct

6) Rank all location types by weighted conversion rate (where does Airport place?)

```

SELECT
    location_type,
    ROUND(SUM(units_sold) * 1.0 / NULLIF(SUM(avg_daily_footfall),
0), 4) AS conversion_rate
FROM challan_test1
GROUP BY location_type
ORDER BY conversion_rate DESC;
Answer (top to bottom): correct

```

1. Art/Design District = **2.0922**
2. Department Store Atrium = **1.6311**
3. Luxury Mall = **1.4180**
4. **Airport Duty-Free = 1.2391**
- 5 High-Street = **1.0341**

7) What are total units and total footfall for Airport Duty-Free?

```

SELECT
    SUM(units_sold) AS total_units_sold,
    SUM(avg_daily_footfall) AS total_footfall
FROM challan_test1
WHERE location_type = 'Airport Duty-Free';

```

Answer: Total Units = **856,913**, Total Footfall = **691,581** correct

8) Average units sold: Airport vs Non-Airport

```

SELECT
    CASE WHEN location_type = 'Airport Duty-Free' THEN 'Airport'
    ELSE 'Non-Airport' END AS location_group,
    ROUND(AVG(units_sold), 2) AS avg_units_sold
FROM challan_test1
GROUP BY 1;

```

Answer: Airport = **1974.45**, Non-Airport = **1927.94** correct

9) Average sell-through %: Airport vs Non-Airport

```

SELECT
    CASE WHEN location_type = 'Airport Duty-Free' THEN 'Airport'
    ELSE 'Non-Airport' END AS location_group,
    ROUND(AVG(sell_through_pct), 2) AS avg_sell_through_pct
FROM challan_test1
GROUP BY 1;

```

Answer: Airport = **73.82%**, Non-Airport = **73.08%** correct

10) In Airport Duty-Free, which regions have the best weighted conversion rate?

```

SELECT
    region,
    COUNT(*) AS events,
    ROUND(SUM(units_sold) * 1.0 /
NULLIF(SUM(avg_daily_footfall),0), 4) AS weighted_conversion_rate
FROM challan_test1
WHERE location_type='Airport Duty-Free'
GROUP BY region
ORDER BY weighted_conversion_rate DESC;
Answer (Airport only):

```

- North America (81) = **1.3216**
- Latin America (92) = **1.2950**
- Asia-Pacific (88) = **1.2356**
- Europe (80) = **1.2015**
- Middle East (93) = **1.1551**

11) Top 5 Airport cities by weighted conversion rate (min 10 events)

```

SELECT
    city,
    COUNT(*) AS events,
    ROUND(SUM(units_sold) * 1.0 /
NULLIF(SUM(avg_daily_footfall),0), 4) AS weighted_conversion_rate
FROM challan_test1
WHERE location_type='Airport Duty-Free'
GROUP BY city
HAVING COUNT(*) >= 10
ORDER BY weighted_conversion_rate DESC
LIMIT 5;
Answer:

```

1. Lima (17) = **1.6519**
2. Paris (14) = **1.5584**
3. Los Angeles (16) = **1.4117**

4. New York (15) = **1.4027**

5. Madrid (21) = **1.3598**

****CTE: Airport vs Non-Airport KPI (events, units, footfall, weighted + avg conversion)**

```
WITH base AS (  
    SELECT  
        CASE WHEN location_type = 'Airport Duty-Free' THEN 'Airport'  
        ELSE 'Non-Airport' END AS location_group,  
        units_sold,  
        avg_daily_footfall,  
        (units_sold * 1.0) / NULLIF(avg_daily_footfall, 0) AS conv  
    FROM challan_test1  
)  
SELECT  
    location_group,  
    COUNT(*) AS events,  
    SUM(units_sold) AS total_units,  
    SUM(avg_daily_footfall) AS total_footfall,  
    ROUND(SUM(units_sold) * 1.0 /  
    NULLIF(SUM(avg_daily_footfall),0), 4) AS weighted_conv,  
    ROUND(AVG(conv), 4) AS avg_event_conv  
FROM base  
GROUP BY location_group;  
Answer
```

- Airport: **434** events, **856,913** units, **691,581** footfall, weighted_conv **1.2391**, avg_event_conv **1.3625**
- Non-Airport: **1,699** events, **3,275,571** units, **2,309,776** footfall, weighted_conv **1.4181**, avg_event_conv **1.6745**

5. High-Street **1.0341**

*** Self-join: Cities where Airport conversion beats Non-Airport (min 5 events each)**

```

WITH city_conv AS (
    SELECT
        city,
        CASE WHEN location_type='Airport Duty-Free' THEN 'Airport'
        ELSE 'Non-Airport' END AS grp,
        COUNT(*) AS events,
        SUM(units_sold) * 1.0 / NULLIF(SUM(avg_daily_footfall),0) AS
weighted_conv
    FROM challan_test1
    GROUP BY city, grp
),
a AS (SELECT * FROM city_conv WHERE grp='Airport'),
n AS (SELECT * FROM city_conv WHERE grp='Non-Airport')
SELECT
    a.city,
    a.events AS airport_events,
    n.events AS non_airport_events,
    ROUND(a.weighted_conv, 4) AS airport_conv,
    ROUND(n.weighted_conv, 4) AS non_airport_conv,
    ROUND(a.weighted_conv - n.weighted_conv, 4) AS diff
FROM a
JOIN n ON a.city = n.city
WHERE a.events >= 5 AND n.events >= 5
ORDER BY diff DESC
LIMIT 5;
Answer (top 5 cities where Airport > Non-Airport)

```

- Lima: **1.6519** vs **1.3629** (diff **+0.2890**)
- Paris: **1.5584** vs **1.4362** (diff **+0.1222**)
- New York: **1.4027** vs **1.2848** (diff **+0.1179**)
- Tokyo: **1.2696** vs **1.1932** (diff **+0.0764**)
- Los Angeles: **1.4117** vs **1.3608** (diff **+0.0510**)

QUERIES FOR PRODUCT ANALYSIS

```

select product_name , (units_sold * price_usd) as total_revenu, units_sold, avg_daily_footfall
from public.challan_test1
order by total_revenu desc limit 10;

```



```
select product_name , (units_sold * price_usd) as total_revenu, units_sold, avg_daily_footfall
from public.challan_test1
order by units_sold desc limit 10;
```

```
select product_name , (units_sold * price_usd) as total_revenu, units_sold, avg_daily_footfall
from public.challan_test1
order by avg_daily_footfall desc limit 10;
```

```
select count(product_name), sum(units_sold * price_usd) as total_revenu,
case when product_name like '%Fragrance' then 'Fragrance' else 'Make-up/skincare' end as
ProductType
from public.challan_test1
group by ProductType;
```

```
select brand, (units_sold * price_usd) as total_revenu
from public.challan_test1
order by total_revenu desc
limit 10;
```

```
select brand, (units_sold * price_usd) as total_revenu, sum(avg_daily_footfall )
from public.challan_test1
group by brand, total_revenu
order by total_revenu desc
limit 10;
```

```
select brand, (units_sold * price_usd) as total_revenu, sum(avg_daily_footfall ) as daily_footfall
from public.challan_test1
group by brand, total_revenu
order by daily_footfall asc
limit 10;
```

THANK YOU.